LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Importance of Calibration, classification of measurements, general care of instruments, objective of metrology, Standardization and various
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Importance of Calibration, classification of measurements, general care of instruments, objective of metrology, Standardization and various Standardizing organizations.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of calibration procedure on standard gauge; Safety: Handle glass/ceramic standards with care; Supervision: Instructor monitors handling of precision tools.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of calibration in engineering?
(b) Name one reason why instrument accuracy matters in aviation maintenance.
(c) Define the term 'metrology' in your own words.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Importance of Calibration • Classification of Measurements • General Care of Instruments • Objective of Metrology • Standardization Concepts	L & Ds	PPT, IB	15 min
(b)	• Role of Standardizing Organizations • Traceability Hierarchy • Common Measurement Errors • Preventive Maintenance Schedules • Documentation Requirements	L & Ds	PPT, PDF	20 min
(c)	• Practical Calibration Demonstration • Instrument Handling Protocols • Error Identification Exercise • Record Keeping Best Practices • Safety Precautions Review	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between accuracy and precision in the context of calibration.
(b) Describe the role of a National Standards Laboratory in the traceability chain.
(c) List three specific steps for general care of precision measuring instruments.

12. Home Assignment and Follow Up	Prepare a brief report on the standardizing organizations relevant to your specific engineering trade, focusing on their role in maintaining measurement standards.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Precision measurements, characteristic of precision measuring instruments.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min